

Storage, File Systems & Forensic Basics

How Storage Works

A disk is a block array. The file system organizes blocks into:

- Files (inodes/mft entries)
- Directories (trees)
- Free space (unallocated blocks)

Deleting a File Does Not Delete Data

When Windows "deletes" a file:

- The file's directory entry is marked free
- The disk blocks are NOT erased – the data remains
- Only the pointers are removed

This is why digital forensics can "recover deleted files" – the bytes are still sitting on the disk in **unallocated space**.

Forensic golden rule: The filesystem leaves artifacts everywhere. Deleting, formatting, and even rebuilding rarely destroy everything. Wiping (overwriting multiple passes) or physical destruction is the only reliable erasure.

Key Forensic Artifacts

Artifact	Location	What It Reveals
\$MFT / inode table	NTFS / ext4	Every file ever created, even deleted
Prefetch	C:\Windows\Prefetch	Programs that ran
\$LogFile	NTFS	File operation history

Artifact	Location	What It Reveals
Shell bags	Registry	Folders user opened in Explorer
Browser history	AppData	Web activity
Pagefile/hiberfile	C:\pagefile.sys	RAM that spilled to disk

The Forensic Process (First Look)

Preserve – image the disk byte-for-byte, write-protected

Document – hashes (SHA-256/SHA-1) to prove integrity

Analyze – study the image, never the original

Recover – carve deleted data

Report – present findings in court or incident report

The first step of any investigation is preservation. If you touch the evidence, you may destroy it. Forensics runs on **images**, not originals.

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